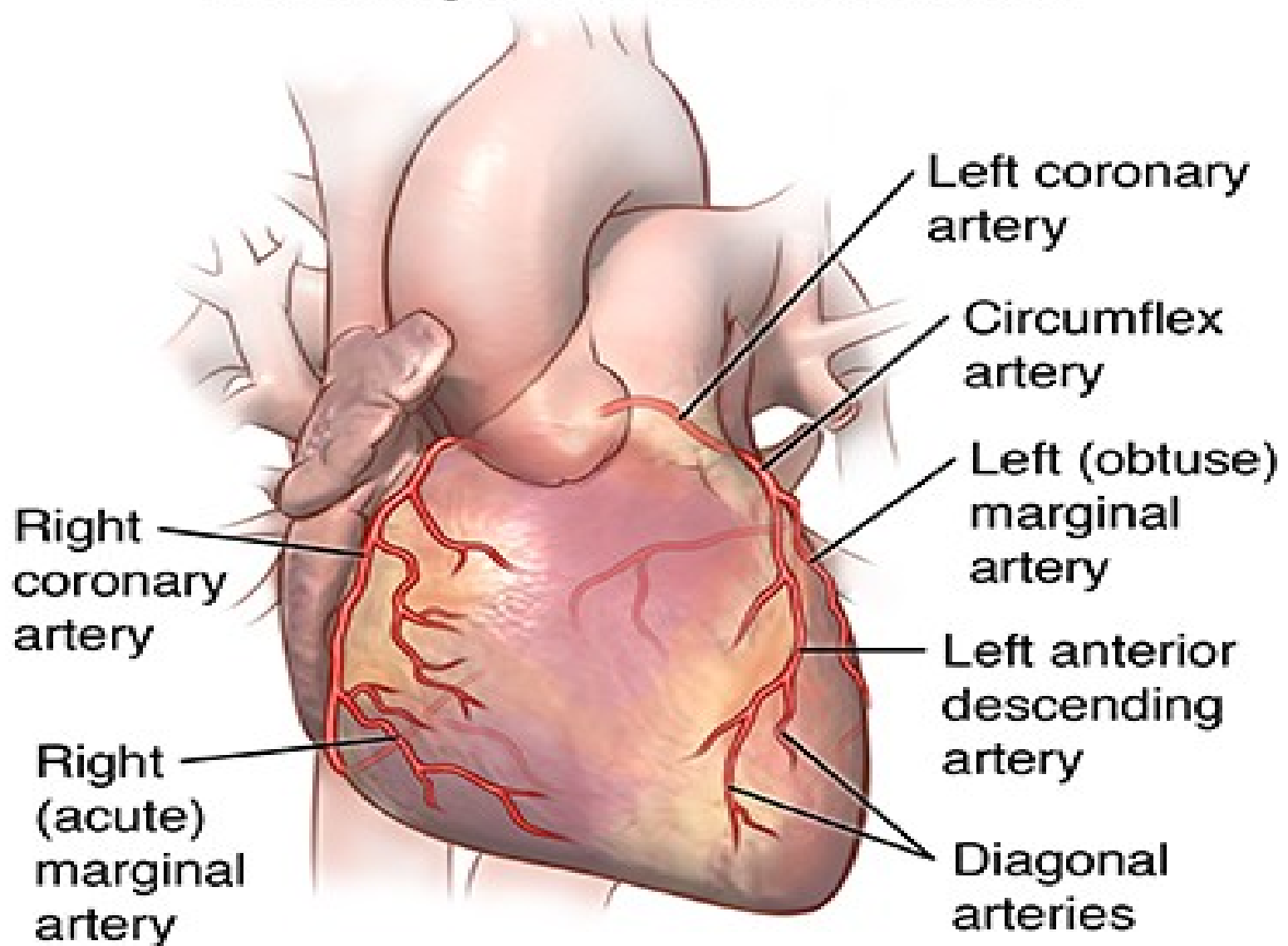

Coronary Artery Disease

Introduction

Coronary arteries of the heart



... ***Intro***

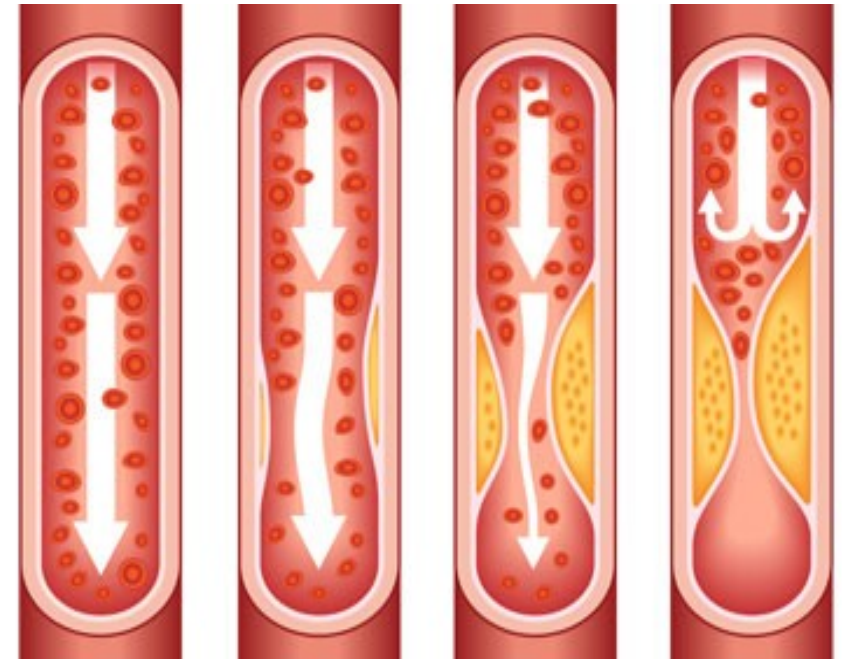
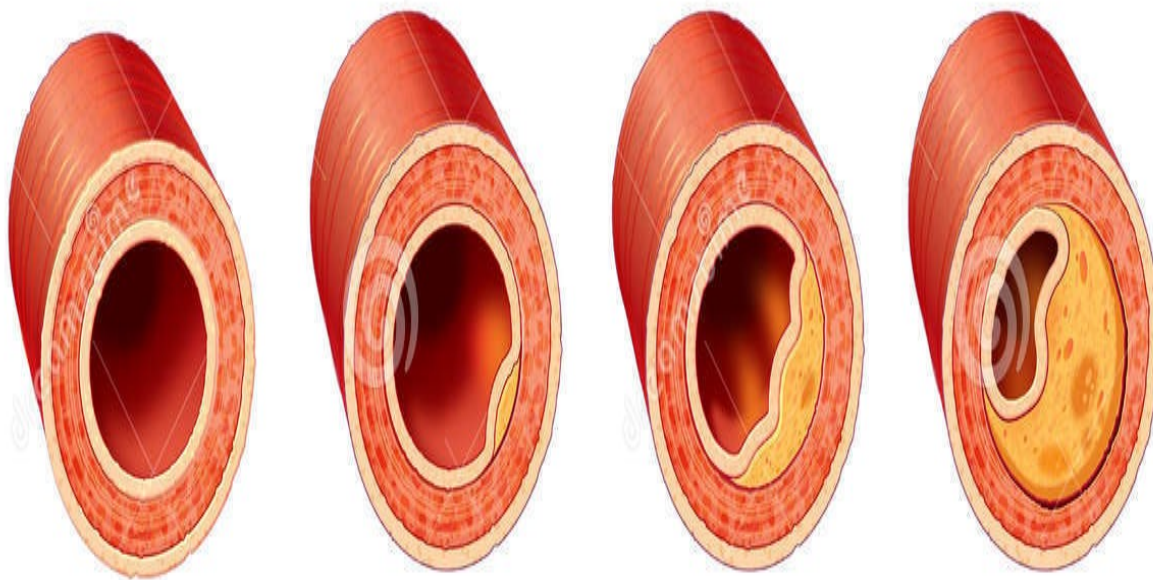
- ☑ **Coronary Heart Disease (CHD)**
- ☑ **Coronary Artery Disease (CAD)**
- ☑ **Ischemic Heart Disease (IHD)**

***Similar
terminologies***

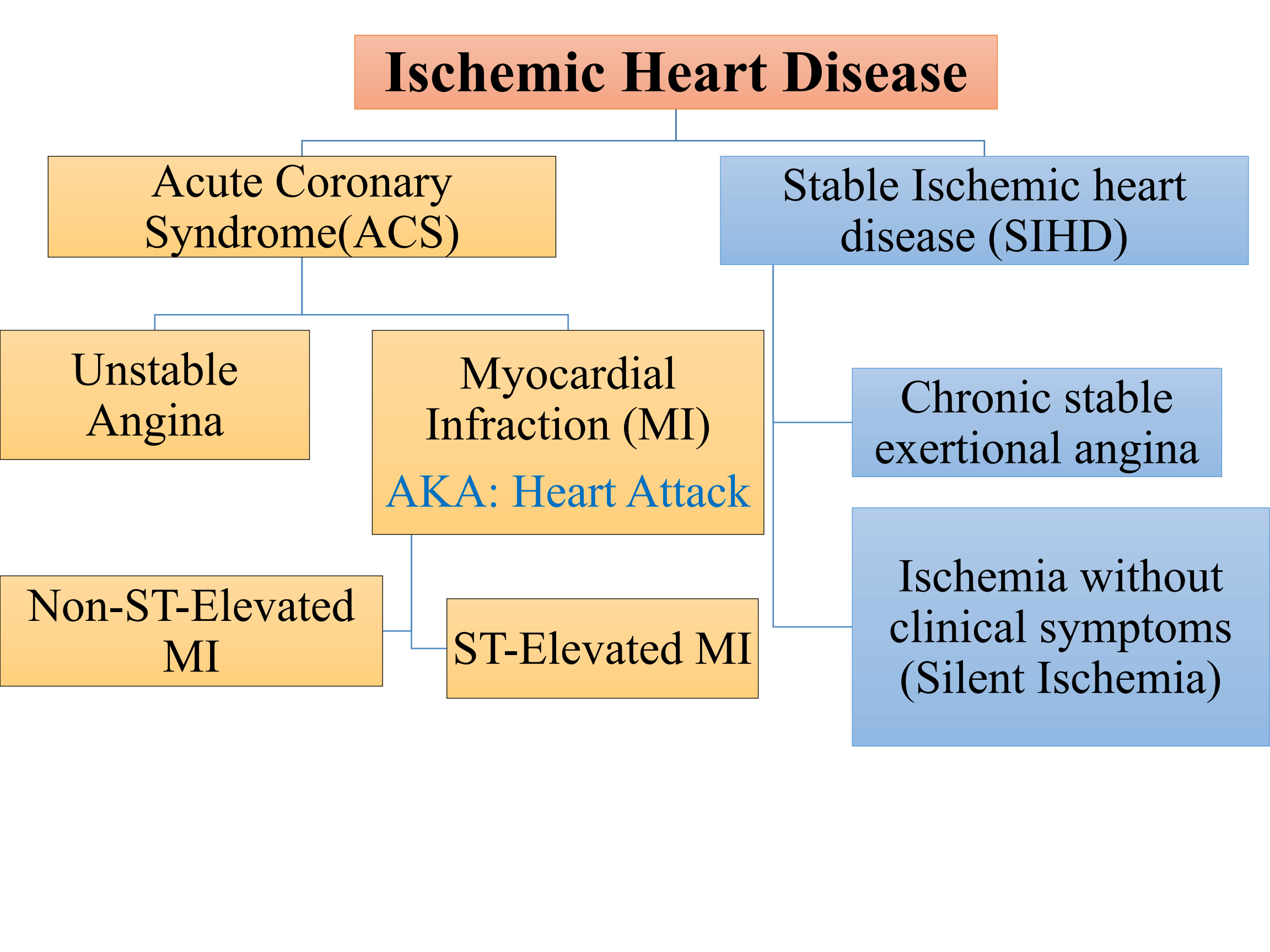
☞ Is **lack of O₂** and **decreased or no blood flow** to the myocardium
resulting from **coronary artery narrowing or occlusion.....**
mostly by atherosclerotic plaque formation

☞ Leads to imbalance between myocardial **O₂ supply & demand**

- The reduction in coronary blood flow>myocardial ischemia
- If left untreated, may lead to myocardial infarction (MI)



Ischemic Heart Disease



```
graph TD; A[Ischemic Heart Disease] --> B[Acute Coronary Syndrome(ACS)]; A --> C[Stable Ischemic heart disease (SIHD)]; B --> D[Unstable Angina]; B --> E["Myocardial Infraction (MI)  
AKA: Heart Attack"]; E --> F[Non-ST-Elevated MI]; E --> G[ST-Elevated MI]; C --> H[Chronic stable exertional angina]; C --> I["Ischemia without clinical symptoms  
(Silent Ischemia)"];
```

The diagram is a hierarchical flowchart. At the top is a box labeled 'Ischemic Heart Disease'. A line from this box branches into two boxes: 'Acute Coronary Syndrome(ACS)' on the left and 'Stable Ischemic heart disease (SIHD)' on the right. From 'Acute Coronary Syndrome(ACS)', a line branches into 'Unstable Angina' and 'Myocardial Infraction (MI) AKA: Heart Attack'. From 'Myocardial Infraction (MI) AKA: Heart Attack', a line branches into 'Non-ST-Elevated MI' and 'ST-Elevated MI'. From 'Stable Ischemic heart disease (SIHD)', a line branches into 'Chronic stable exertional angina' and 'Ischemia without clinical symptoms (Silent Ischemia)'.

Acute Coronary
Syndrome(ACS)

Unstable
Angina

Non-ST-Elevated
MI

Myocardial
Infraction (MI)
AKA: Heart Attack

ST-Elevated MI

Stable Ischemic heart
disease (SIHD)

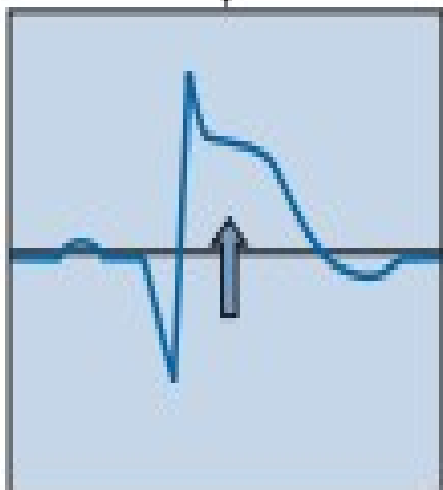
Chronic stable
exertional angina

Ischemia without
clinical symptoms
(Silent Ischemia)

Ischemic chest discomfort symptoms, lasting at least 20 min;
suspect acute coronary syndrome

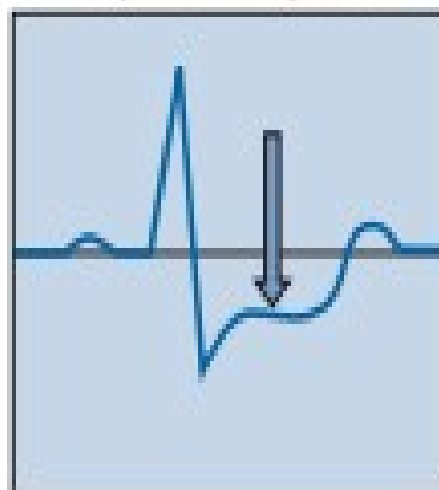
Obtain and interpret a 12-lead ECG within 10 min

ST-segment elevation

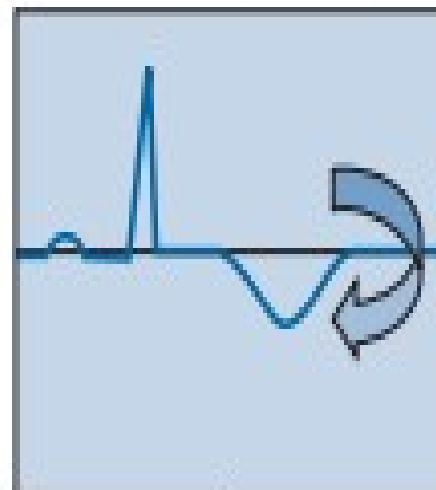


No ST-segment elevation

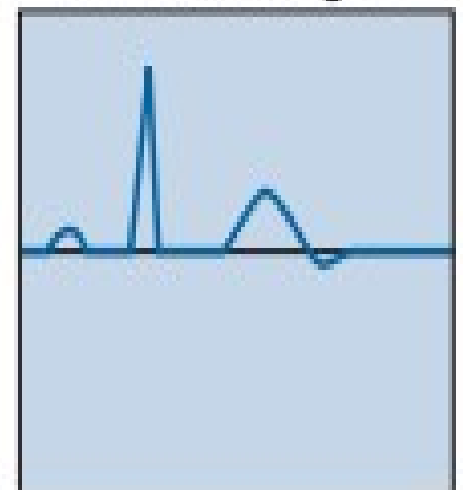
ST-segment depression

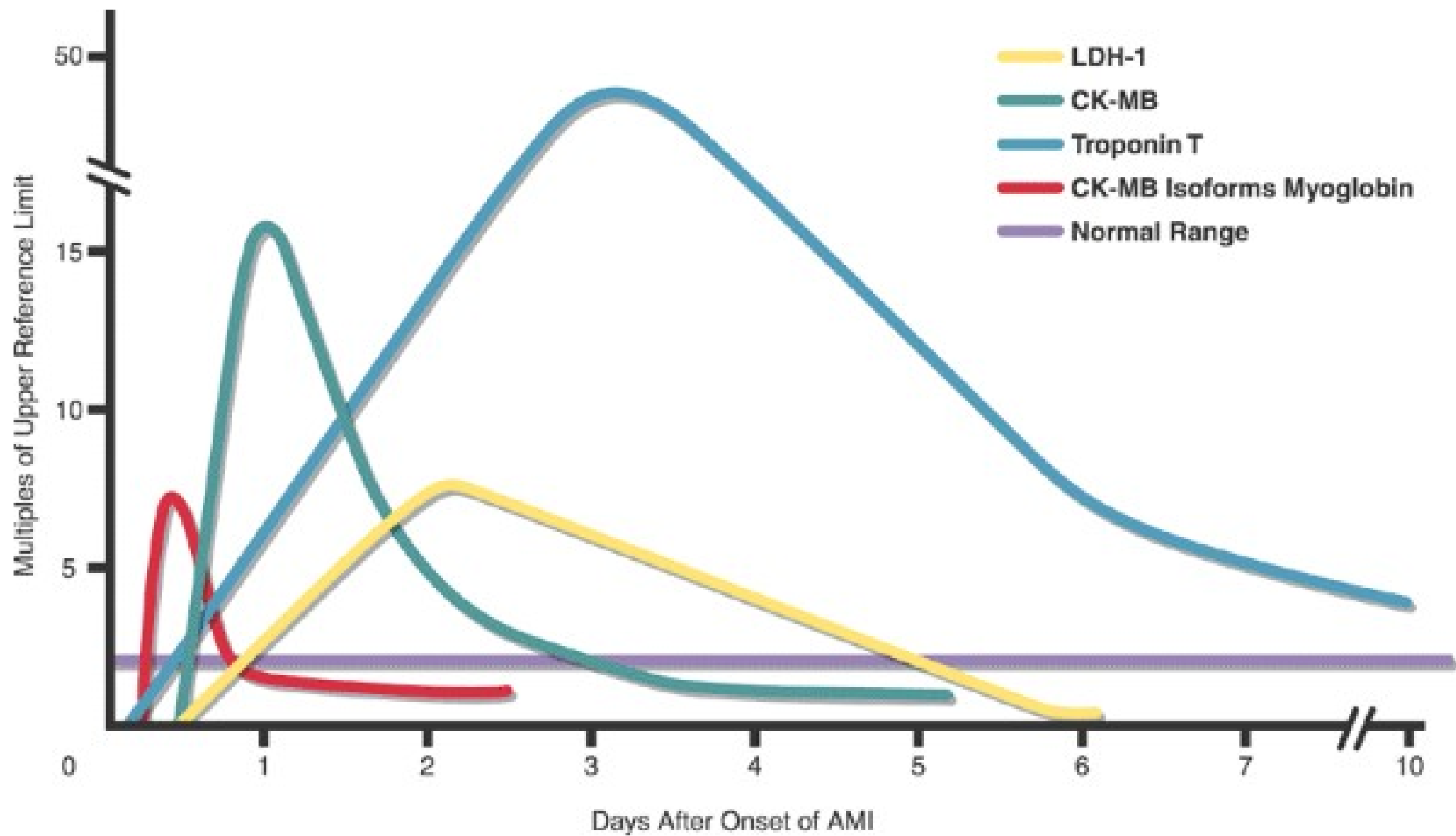


T-wave inversion



No ECG changes





Comparison of

	<i>Unstable Angina</i>	<i>NSTEMI</i>	<i>STEMI</i>
1. Sxs present	+	+	+
2. ECG changes	• None	• ST segment depression, • T wave inversion, or • No changes	• ST segment elevation
3. Other waves	• Non-Q waves	• Non-Q waves	• Pathologic Q waves • New left bundle branch block
4. Biochemical marker	• No changes	• ↑ troponins & CK MB	• ↑↑↑ troponins & CK MB
5. Occlusion	• Non-occlusive thrombus	• Partial occlusive thrombus	• Total occlusive thrombi
6. Spectrum of damage		• Involves only sub-endocardial myocardium	• Extensive damage & injury to entire thickness of myocardium

Epidemiology

☑ According to the latest WHO data published in may 2014,

↪ CHD deaths in Ethiopia reached 14,728 or 2.45% of total deaths.

↪ The age adjusted death rate is 36.10 per 100,000 of population ranks

↪ CHD is the 6th leading cause of death in Ethiopia

Etiology

Atherosclerosis plaque

- **Endothelial dysfunction, Inflammation, and fatty streaks**

Vasospasm

 Platelet aggregation, cocaine abuse

Emboli

 AF, vegetation of infective endocarditis

Risk Factors for CAD

I. Non- modifiable

- a. **Advanced age**
 - ≥ 55 and ≥ 65 yrs for male & female respectively
- b. **Male**
- c. **Family Hx** of early onset CVD (premature CVD)

N.B. The presence of any RF is associated with doubling the risk of an MI.

II. Modifiable

- a. **Hypertension:** Due to shear stress
- b. **Diabetes Mellitus**
- c. **Dyslipidemia**
- d. **Stress:**
- e. **Obesity** ($\text{BMI} \geq 30 \text{ kg/m}^2$)
- f. **Albuminuria**
- g. **Physical inactivity**
- h. **Cigarette smoking:**
- i. **Chronic inflammation**
 - i.e., gingivitis, arthritis, elevated c-reactive protein
- j. **Illicit drug use**
 - e.g., cocaine, amphetamine, progestins, corticosteroids

Pathophysiology

Platelet adhesion, activation & aggregation $\Rightarrow \Rightarrow$ Cross-links(

(+) coagulation cascade $\Rightarrow \Rightarrow \Rightarrow$
Thrombus $\Rightarrow$ Fibrin clot

CV Risk Factors

Endothelial injury (streak)

Inflammation & immune reactions

Accumulation of lipids in the intima of arteriolar wall

T-lymphocytes & monocytes infiltrate the area to ingest the lipids & then die

Proliferation of smooth muscle cells within the vessel

Formation of fibrous cap within the dead fatty core = Atheroma

Protrusion of atheroma into the lumen of the vessel

Atherosclerosis

Atherosclerosis with blood clot formation = Atherosclerosis Plaque (AP)

Rapture, fissuring & erosion of unstable AP

Rapture ,
fissuring &
erosion of
unstable AP

Distal
embolization

Occlusion of
coronary
arteries (CAs)

CAs cannot supply
enough blood (O₂,
glucose & nutrients) to the
heart

Myocardial
ischemia (Ischemic
myocardial cells – IMCs)

IMCs may have
decreased electrical
& muscular
function

IMCs convert
to anaerobic
mbm

IMCs produce
**lactic acid as
waste**

Pain develops from
lactic acid
accumulation

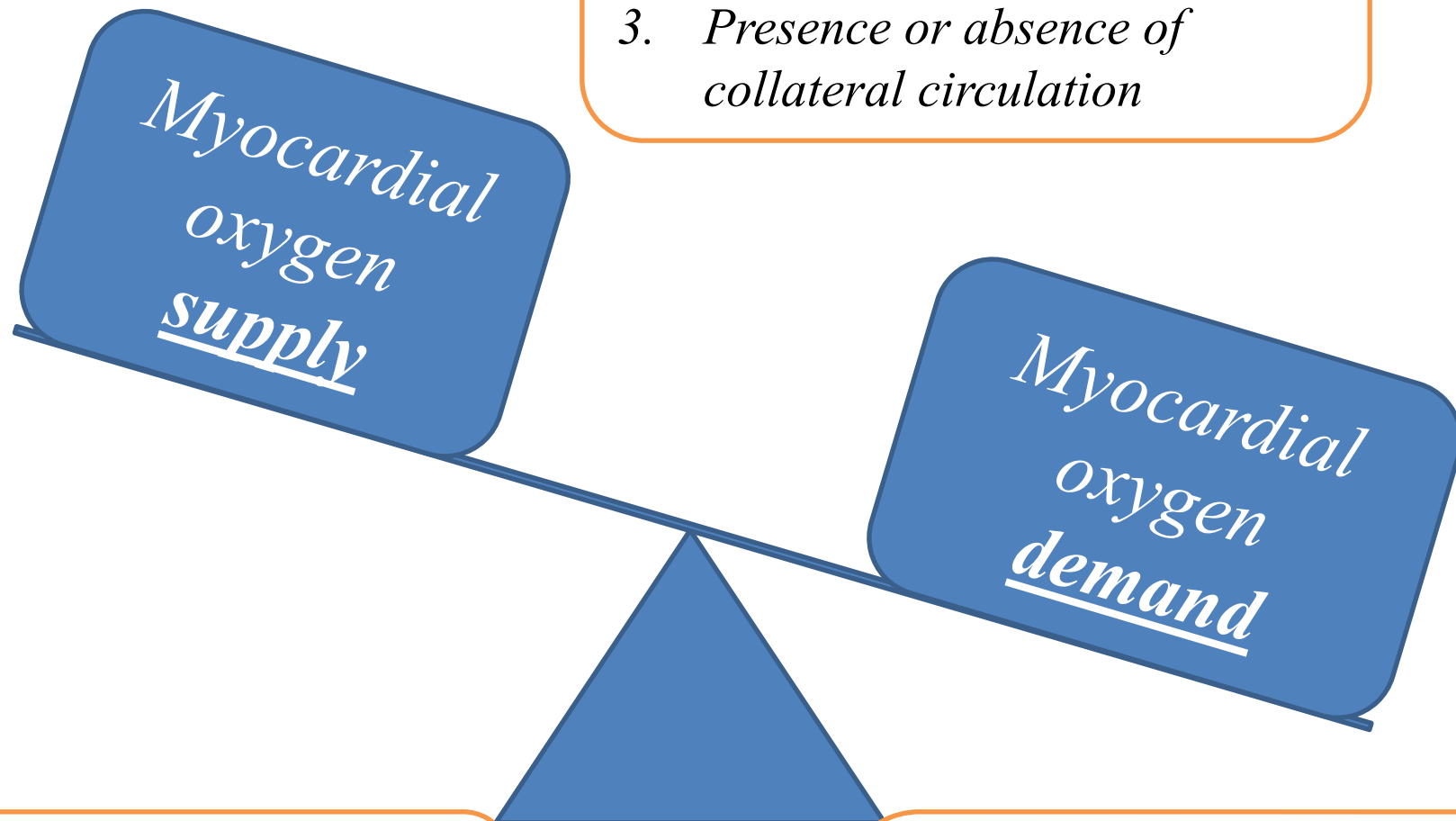
Myocardial
Infarction (Angina
symptoms)

Myocardial
Remodeling

Heart Failure

Severity of MI depends on

1. *Level of occlusion*
2. *Length of time of occlusion*
3. *Presence or absence of collateral circulation*



Determinants of M-O₂-Supply

1. *Coronary blood flow*
2. *Regional myocardial blood flow*

Determinants of M-O₂ -demand

1. *Heart rate*
2. *Contractility*
3. *Intra-myocardial wall tension during systole [preload & afterload]*

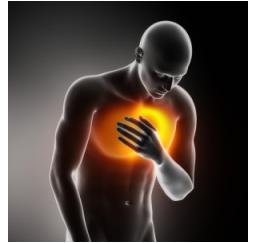
... *Patho*

☑ **Complications of MI**

- ↪ Cardiogenic shock
- ↪ Heart failure (HF)
- ↪ Valvular dysfunction
- ↪ Arrhythmias
- ↪ Pericarditis
- ↪ Stroke 2^o to LV thrombus embolization
- ↪ Venous thromboembolism, and
- ↪ LV free-wall rupture.

Clinical Presentation

✓ A. Midline anterior angina pain/Chest discomfort



↳ The most common initial & predominant CP

↳ Sensation of *pressure, burning, heaviness, fullness or crushing sensation* **over or near sternum**; often but not always radiating to *left jaw, shoulder, arm, back*

↳ **Precipitating factors:**

- Exercise, cold environment, walking after a meal,
- Emotional upset, fright, anger, coitus

✓ B. Accompanying symptoms

↳ NV, shortness of breath, diaphoresis, cool sweat, $\uparrow T^0$

- **For ACS:** Pain is not relieved
- **For chronic angina:** Pain is relieved

[By rest, pain medications, nitrates or position changes]

DIAGNOSIS

↪ A 12-lead ECG should be obtained within 10 minutes of pt presentation.

↪ Key findings indicating myocardial ischemia or MI are

✓ **ST-segment** elevation: STEMI

✓ ST-segment depression

✓ T-wave inversion



NSTEMI

DIAGNOSIS

- ↪ The appearance of a new left bundle-branch block accompanied by chest discomfort is highly specific for acute MI.
- ↪ Some pts with myocardial ischemia have no ECG changes, so biochemical markers and other risk factors for CAD should be assessed

DIAGNOSIS

- ↳ Biochemical markers are important for confirming the diagnosis of MI.
- Typical rise and gradual fall in **troponin I** or **T**
- More rapid rise and fall of CK-MB
- ↳ Both troponins and CK-MB are detectable within 2-4 hrs of MI.
- ↳ Troponins remain elevated for up to >7 days, whereas CK-MB returns to normal within 48 hrs

STEMI vs. NSTEMI

STEMI

- ♠ Totally occlusive thrombus
- ♠ Extensive damage
- ♠ Injury to entire thickness of myocardium (**transmural**)
- ♠ Occlusion leads to myocardial necrosis
- ♠ ST segment elevation
- ♠ Primary reperfusion or thrombolytic therapy

Non-STEMI

- ⌘ Partially occlusive thrombus
- ⌘ Less extensive damage
- ⌘ Involves only **sub-endocardial myocardium**
- ⌘ ST depression or no ST elevation
- ⌘ Anti-platelet agents used to target thrombus

∞ Blood chemistry tests

- K & Mg, glucose, SCr, baseline CBC count and coagulation tests & fasting lipid panel.

∞ It is important to triage and treat pts according to their risk category.

∞ Pts with STEMI are at high risk of death and efforts to reestablish coronary perfusion should be initiated immediately

Treatment

A. Short Term Goals

- (1) **Early restoration** of blood flow to the infarct-related artery to prevent infarct expansion (in the case of MI) or prevent complete occlusion and MI (in UA)
- (2) Prevention of **death and other complications**
- (3) Prevention of **coronary artery re-occlusion**
- (4) Relief of ischemic **chest discomfort**, and
- (5) Resolution of **ST-segment and T-wave changes** on ECG

... ***Rx***

B. Long-term goals

 Prevent additional CHD events

- MI
- Arrhythmias
- Heart failure

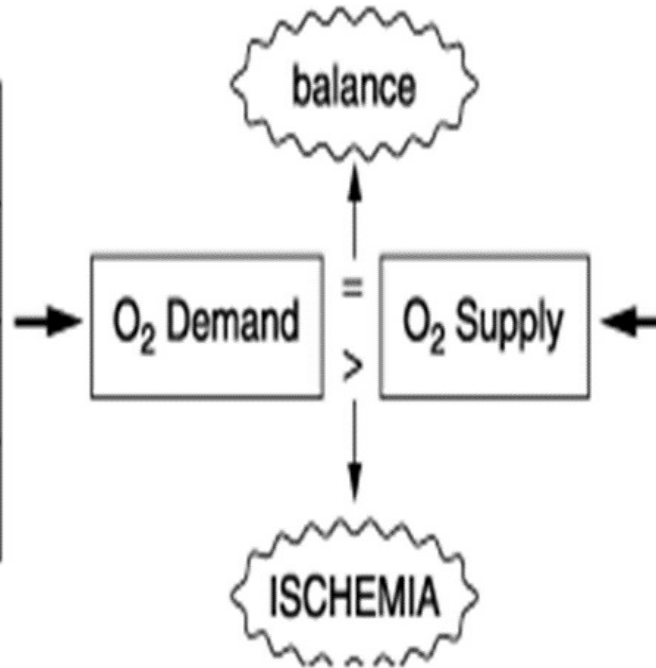
 Extend the patient's life

 Improve the patient's QOL

Agents decreasing O₂ Demand

β adrenergic antagonists
some Ca²⁺ entry blockers

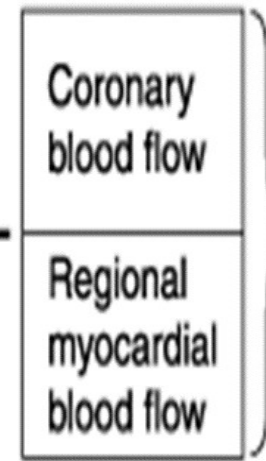
Organic nitrates
Ca²⁺ entry blockers



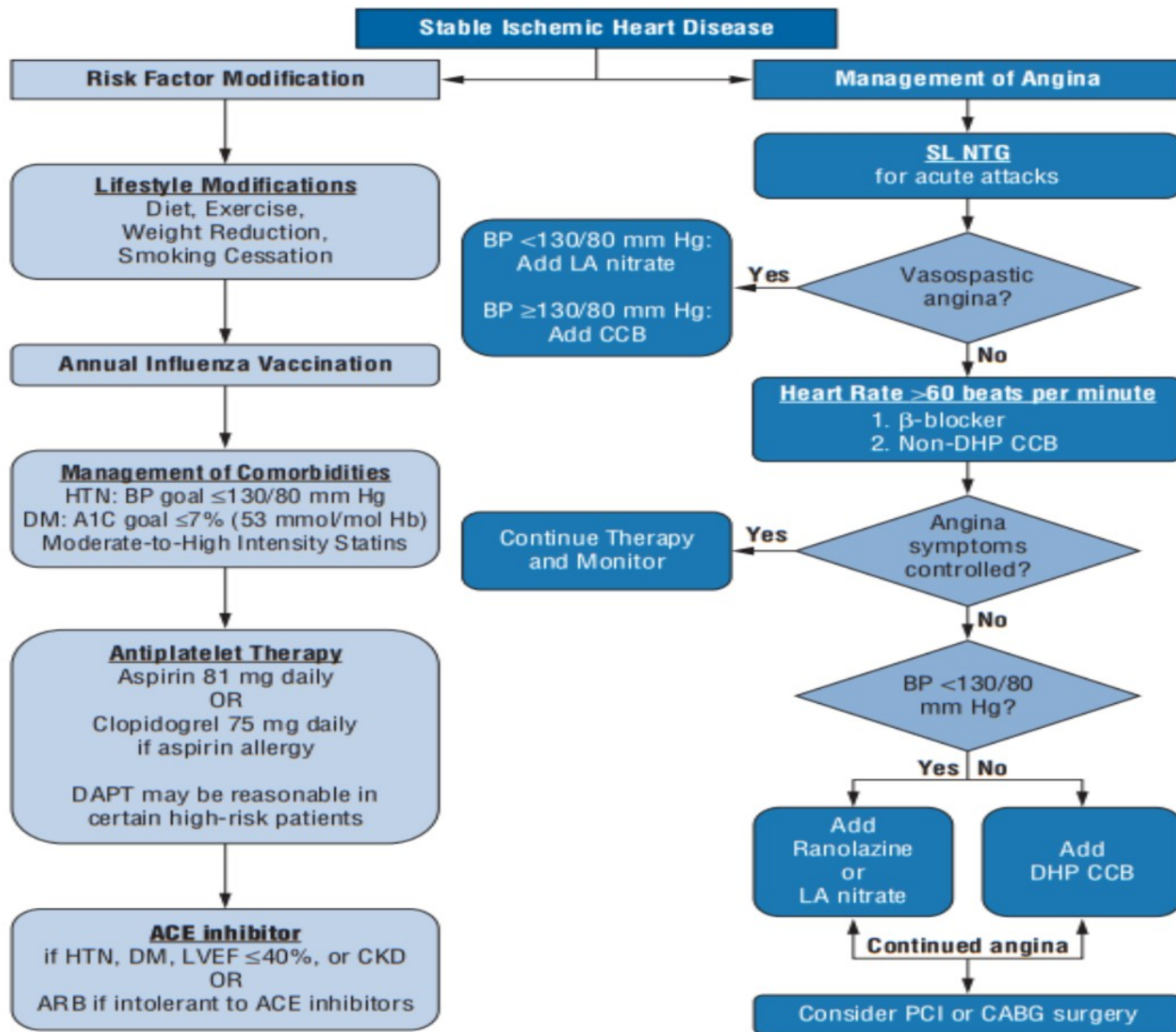
Agents increasing O₂ Supply

Vasodilators
(esp. Ca²⁺ entry blockers)

Also: statins,
anti-thrombotics



<i>Medications</i>	<i>Decrease O₂ Demand</i>	<i>Increase O₂ Supply</i>
<i>CCB</i>	☑	☑
<i>B-blockers</i>	☑	☒
<i>Nitrates</i>	☑	☑
<i>Statins</i>	?	☑
<i>Antithrombotics</i>	?	☑



Early Rx of ACS

1. MONA-B

A. **M**orphine [for selected patients]

B. **O**₂: If O₂ saturation is low (< 90%)

C. **N**itroglycerine: 0.3-0.4 mg SL every 5 minutes as needed, up to 3 doses

D. Aspirin: 162-325 mg (chew & swallow) [Start within 24 hours] [To all patients]

D. B-Blockers:

2. Platelet P2Y₁₂ receptor inhibitor: Clopidogrel, Prasugrel: For at least 12 months

4. Glycoprotein IIb/IIIa Receptor Inhibitors: Eptifibatide, and tirofiban

4. Anticoagulants: For the duration of hospitalization & for at least 8 hrs

4. Stool softeners: E.g., bisacodyl

4. ACEIs : [Start within 24 hours]

4. CCBs

4. Statins: High dose /High intensity

Early STEMI treatment strategies

- ☒ Primary PCI within 90 minutes-1st choice
- ☒ Fibrinolysis within 120 minutes, if no PCI
 - ✓ Alteplase, Reteplase, Tenecteplase, Streptokinase
- ☒ Antiplatelet
- ☒ Antithrombotic

... **Rx**

Contraindications to Fibrinolysis

Absolute contraindications:

1. Active internal bleeding (not including menses)
2. Previous intracranial hemorrhage (Hx of hemorrhagic stroke) at any time
3. Ischemic stroke within 3 months
4. Known intracranial neoplasm
5. Known structural CV lesion (e.g., Arteriovenous malformation)
6. Suspected aortic dissection
7. Significant closed head or facial trauma within 3 months

... **Rx**

 Relative contraindications:

- Uncontrolled HTN (BP > 180/110 mm hg)
- Ischemic stroke > 3 months
- Dementia
- Intracranial pathology
- Current anticoagulant use
- Bleeding diathesis
- Traumatic or prolonged CPR (> 10 min)
- Major surgery (< 3 wks)

... **Rx**

Avoid [not recommended]

 NSAIDs & COX-2 inhibitors

- B/c they increase the risk of death, infarction, HF & myocardial rupture

 Hormone therapy (estrogen/progestin):

- B/c they increase risk of CV events

 Garlic.....bleeding risk

Antiplatelet Therapy

T.M.



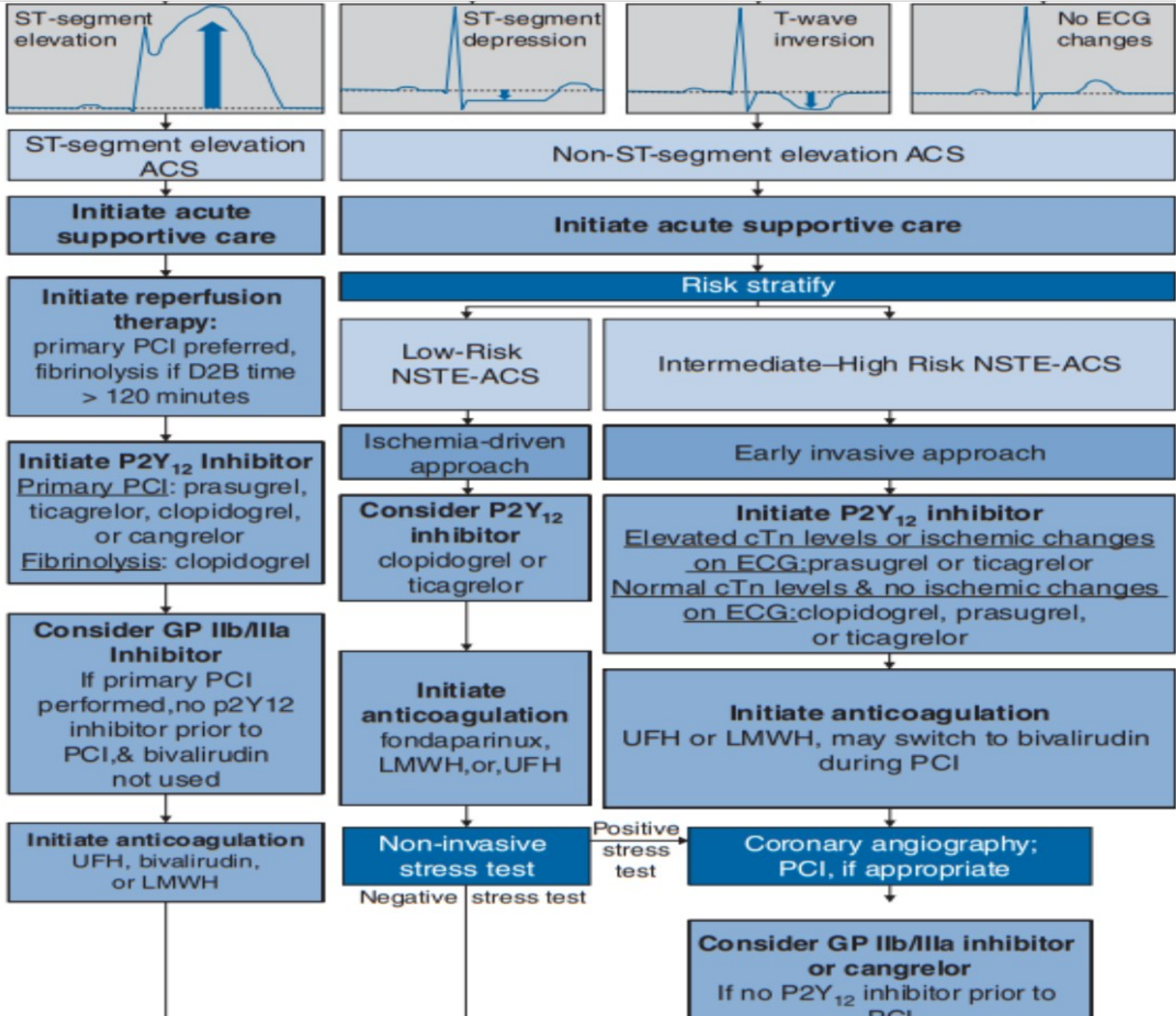
T.M.

- ☑ **Aspirin:** A dose of 81 mg/day continued indefinitely
- ☑ **P2Y12 Inhibitors:** given with aspirin as dual antiplatelet therapy (DAPT) for 12 months
- ✓ **Clopidogrel:** 600-mg oral LD before PCI or 300-mg with fibrinolytic or no reperfusion; 75 mg MD
- ✓ **Prasugrel:** 60-mg LD, 10mg MD for ≥ 60 kg or 5mg for < 60 kg pt
- ✓ **Ticagrelor:** 180-mg LD, followed by 90 mg orally twice daily
- ✓ **Cangrelor:** 30 mcg/kg IV bolus, followed by 4 mcg/kg/min infusion

Anticoagulants



- ☒ **UFH:** 70-100 units/Kg for PCI or
60units/Kg IV bolus, followed by
12units/Kg/hr
- ✓ Monitored with the aPTT & HIT
- ☒ **LMWHs:** Enoxaparin 30mg IV bolus
followed by 1mg/kg every 12 hr



Chronic Rx and chronic prophylaxis [Chronic prophylactic Rx]

1. Antiplatelets: Aspirin [Indefinitely] $\pm$ Clopidogrel [At least 12 months]
2. β -blockers: [For at least 3 yrs or Indefinitely for HFrEF].
3. ACEIs: [Indefinitely for **select** patients]
4. CCBs: Alternative to β -blockers
5. Long acting nitrates: Only for infrequent attacks or predictive angina b/c of **tolerance**
6. Statins: [Indefinitely following STEMI or NSTEMI]
7. Annual influenza vaccine

Mgt of tolerance:
Nitrate free interval of 8 hrs/day or longer

Medical Therapy for Relief of Symptoms

1. *β -blockers: [1st line]*
2. *CCBs: [2nd line]*
3. *Long acting nitrates: [3rd line]*
4. *Ranolazine: [4th line]*
5. *Revascularization : [If refractory angina]*

Risk Factor Modification

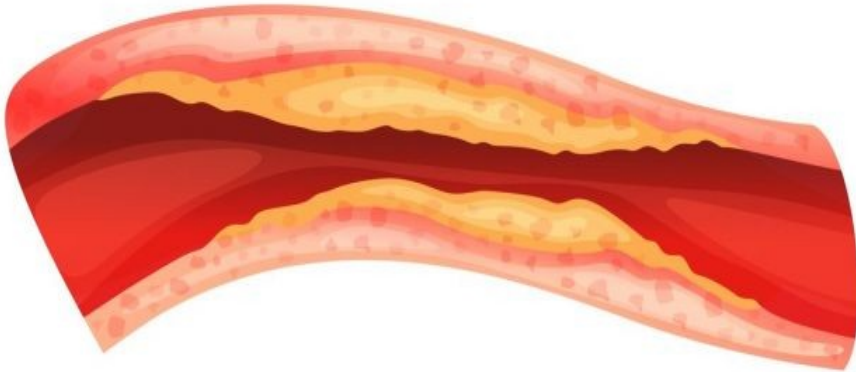
1. *Therapeutic Life style Changes*
2. *Lipid lowering agents*
3. *Antihypertensives*
4. *DM medications*

NONPHARMACOLOGIC THERAPY

- ☑ Risk factor modification
- ☑ Lifestyle modifications:
 - ✓ Physical activity
 - ✓ Weight management
 - ✓ Dietary therapy
 - ✓ Smoking cessation
 - ✓ Psychological interventions (screening and treatment for depression if appropriate)
 - ✓ Limitation of alcohol intake
 - ✓ Avoiding exposure to air pollution.

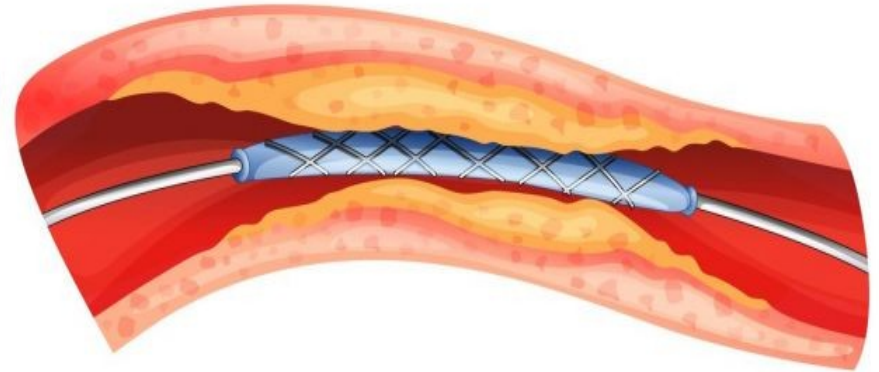
Stent with Balloon Angioplasty

1



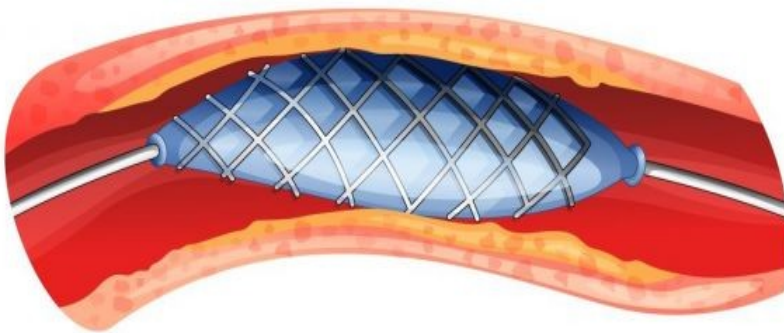
Build up of cholesterol partially blocking blood flow through the artery.

2



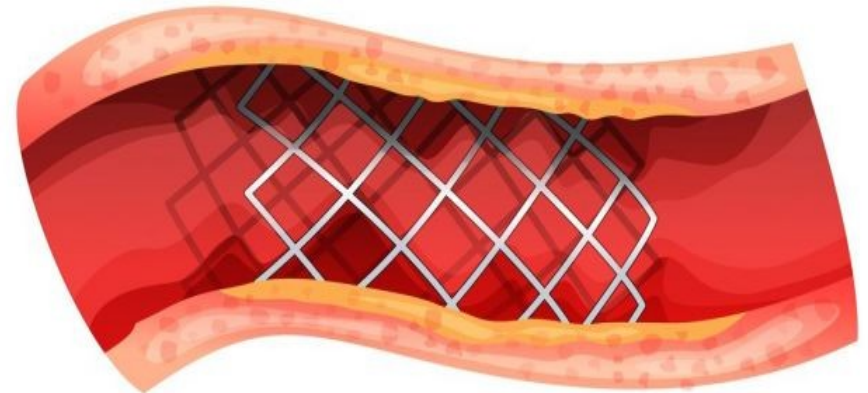
Stent with balloon inserted into partially blocked artery.

3



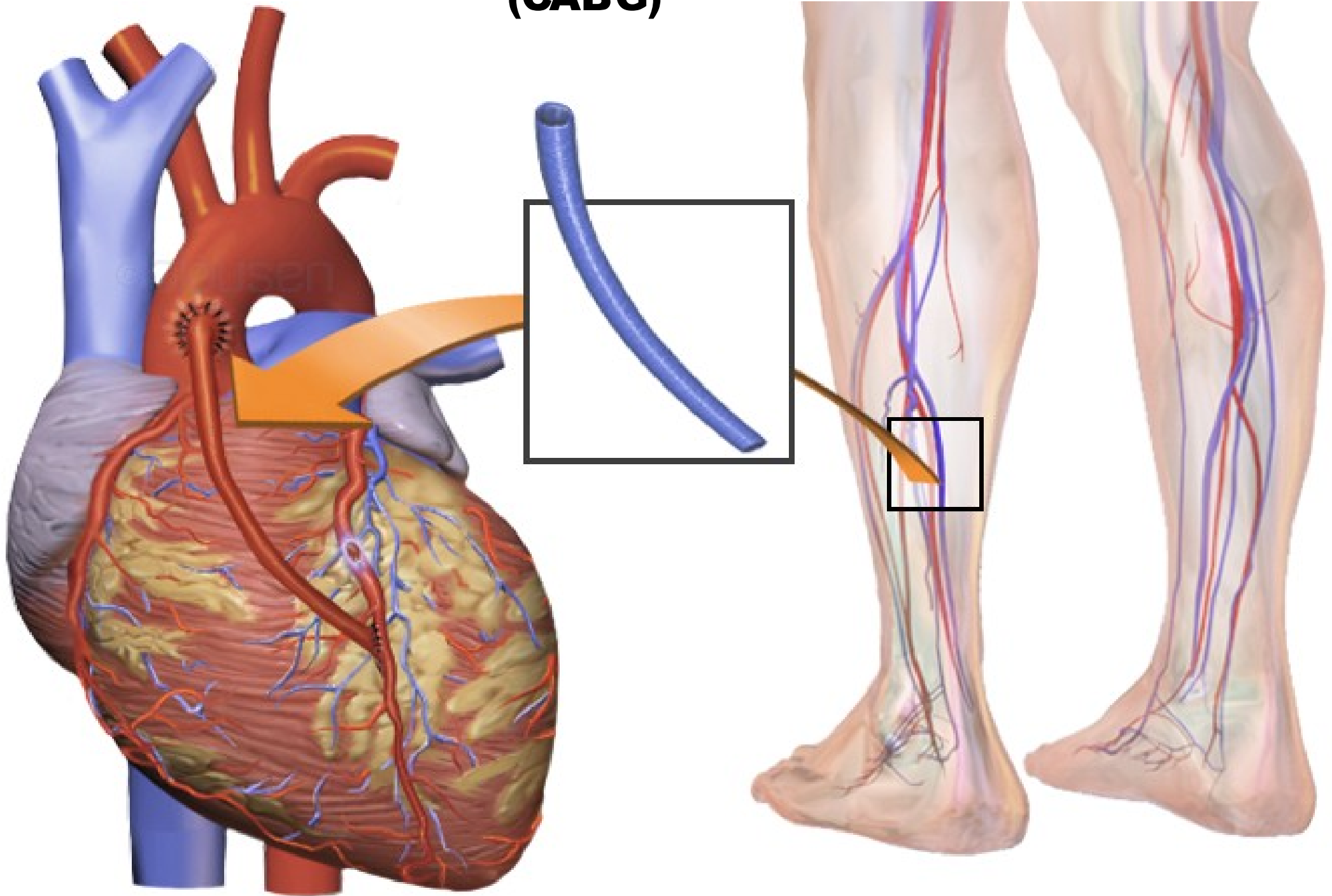
Balloon inflated to expand stent.

4



Balloon removed from expanded stent.

Coronary Artery Bypass Graft (CABG)



7 ***Evaluation of Therapeutic Outcomes***

☒ **A. Subjective data**

↳ Number of painful episodes, amount of rapid-acting NTG consumed, exercise tolerance.

☒ **B. Objective data**

↳ Ischemic changes on ECG

☒ **C. Major AEs**

☒ **D. Ancillary monitoring:** Lipid profiles, FPG, thyroid function tests, Hgb/Hct, and electrolytes.